

نام: علی کاظم نیا شماره دانشجوئی: ۸۸۳۳۲۳۰

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۱- اولا توابع بولی زیر را با استفاده از نقشه کسرتفاده ساده کنید.

4

(الف) $F(x, y, z) = \sum(0, 1, 5, 7)$

5

(ب) $F(x, y, z) = \sum(1, 2, 3, 6, 7)$

6

(ج) $F(x, y, z) = \sum(3, 5, 6, 7)$

7

(د) $F(x, y, z) = \sum(0, 2, 3, 4, 6)$

8

حل:

$F(x, y, z) = \sum(0, 1, 5, 7)$

x \ yz	00	01	11	10
0	1	1	0	0
1	0	1	1	0

(الف) 10

11

12

$F = \bar{x}\bar{y} + xz$

13

14

$F(x, y, z) = \sum(1, 2, 3, 6, 7)$

x \ yz	00	01	11	10
0	0	1	1	1
1	0	0	1	1

(ب) 15

16

17

18

$F = \bar{x}z + y$

19

$F(x, y, z) = \sum(3, 5, 6, 7)$

(ج) 20

21

x \ yz	00	01	11	10
0	0	0	1	0
1	0	1	1	1

22

$F = xz + xy + yz$

2

$$F(x, y, z) = \sum(0, 2, 3, 4, 6) \quad (1)$$

$x \backslash yz$	00	01	11	10
0	1	0	1	1
1	1	0	0	1

$$F = \bar{z} + \bar{x}y$$

9- اولى بولى زير را با استفاده از نقشه چهار متغیره ساده كنيد:

$$F(A, B, C, D) = \sum(4, 6, 7, 15) \quad (الف)$$

$$F(A, B, C, D) = \sum(3, 7, 11, 13, 14, 15) \quad (ب)$$

$$F(A, B, C, D) = \sum(0, 1, 2, 4, 5, 7, 11, 15) \quad (ج)$$

$$F(A, B, C, D) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15) \quad (د)$$

$$F(A, B, C, D) = \sum(4, 6, 7, 15)$$

حل:
(الف)

$AB \backslash CD$	00	01	11	10
00	0	0	0	0
01	1	0	1	1
11	0	0	1	0
10	0	0	0	0

$$F = BCD + \bar{A}B\bar{D}$$

$$F(A, B, C, D) = \sum(3, 7, 11, 13, 14, 15) \quad (ب)$$

$AB \backslash CD$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

$$F = CD + ABD + ABC$$

«3»

$$F(A, B, C, D) = \sum(0, 1, 2, 4, 5, 7, 11, 15) \quad (2)$$

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	1	0
11	0	0	1	0
10	0	0	1	0

$$F = \bar{A}\bar{C} + ACD + \bar{A}\bar{B}\bar{D} + \bar{A}BD$$

$$F(A, B, C, D) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15) \quad (3)$$

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	0	1	1	0
10	1	0	0	1

$$F = BD + \bar{B}\bar{D} + \bar{A}\bar{B}$$

19. F را به دو صورت $F(w, x, y, z)$ و $F(w, x, y, z)$ ساده کنید.
 (2) $F(w, x, y, z) = \sum(0, 2, 2, 3, 7, 8, 10)$
 (3) $F(w, x, y, z) = \sum(5, 6, 11, 15)$
 20. حالات بی اهمیت را به دو صورت $F(w, x, y, z)$ و $F(w, x, y, z)$ ساده کنید.
 (2) $F(w, x, y, z) = \sum(0, 2, 2, 3, 7, 8, 10)$
 (3) $F(w, x, y, z) = \sum(5, 6, 11, 15)$

14n

wx \ yz	00	01	11	10
00	1	1	1	1
01	0	$\pi(1)$	1	$\pi(0)$
11	0	0	$\pi(1)$	0
10	1	0	$\pi(0)$	1

Annotations:

- $\bar{w}z$ (pointing to the 01 column)
- $\bar{w}\bar{z}$ (pointing to the 00 column)
- $\bar{w}z$ (pointing to the 11 column)

~~scribbles~~

5n

A=0

A=1

DE		BC				DE		BC			
		00	01	11	10			00	01	11	10
BC	00	0	1	1	1	BC	00	0	0	1	1
	01	1	1	1	1		01	1	1	1	1
	11	0	1	1	1		11	1	0	1	1
	10	0	0	1	0		10	1	0	1	0

منه صواب (طريق صفر):

$$F = (\bar{A} + B + C + D)(\bar{B} + \bar{A} + D + E)(A + \bar{B} + D + E) \\ (A + \bar{B} + C + D)(A + \bar{B} + C + E)$$

منه صواب (طريق 1):

$$F = \bar{B}\bar{C}D + \bar{B}C + \bar{A}\bar{B}\bar{D}E + BCD + ABD \\ + \bar{A}BDE + \bar{A}BCE$$